MILITARY SPECIFICATION

COMPUTER SET, NAVIGATION

AN/ASN-66(*)

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall compute and display the present position of an aircraft in flight and shall compute the course and distance to a pre-selected destination.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Units	Type Designation	Appl. Para.
Computer-Control	CP-824(*)/ASN-66	3.5.1
Amplifier-Computer	AM-4341(*)/ASN-66	3.5.2

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.8.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-172	Cases, Bases, Mounting; and Mounts, Vibration (for use with Electronic Equipment in aircraft)
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interface Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-S-23603	Systems Readiness/Maintainability; Avionic Systems Design, General Specification for

Naval Air Systems Command

WR-98	Microelectronic Devices Used in Avionic Equipment, Procedures for

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MS17322	Meter, Time Totalizing, 115 Volt, 400-Cycle, Rev. B

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

2.2 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Microelectronic items shall conform to requirements specified herein.

(2) Parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.5)

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in WR-98.

3.2.2 Electron Devices - Transistors and diodes shall be selected and applied as outlined in Specification MIL-E-4682.

3.2.3 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of WR-98.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:35Z
Check the source to verify that this is the current version before use.

3.2.4 Modules, Maintenance - The electronic portions of the equipment shall be divided into maintenance modules (see Specification MIL-S-23603). Maintenance modules shall normally be considered repairable.

3.2.4.1 Module Serviceability - All subassemblies shall be module constructed and shall be interchangeable between equipment and easily removable for maintenance. The number of fasteners having to be removed to remove each module shall be held to a minimum.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 35 pounds.

3.3.2 Reliability -

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 500 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment shall have 150 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 Time Totalizing Meter - The following units shall contain time totalizing meters in accordance with Specification MIL-M-7793.

Unit	Type of Meter
CP-824/ASN-66	MS17322
AM-4341/ASN-66	MS17322

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:35Z
Check the source to verify that this is the current version before use.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all panels must be approved by the procuring activity prior to preproduction testing.

3.3.5 Interchangeability -

3.3.5.1 Interchangeability of Reordered Equipment - For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

3.3.5.2 Modular Interchangeability - All mechanical shaft position scale factors and electrical scale factors shall be constant from one equipment to another and from one subassembly to another such that no adjustments are required to maintain completed interchangeability of identical modules within the units of the Navigation Computer Set, interchangeability of units within the Navigation Computer Set and interchangeability of Navigation Computer Sets.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.3.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C ± 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115 ± 1.0 V AC and/or 27.5 ± 0.5V DC

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment, except as modified herein:

3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of Specification MIL-E-5400, except that the acceleration shall be maximum ± 5G from 40 to 400 cycles.

3.3.11 Warm-Up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 2 minutes under standard conditions and 5 minutes at extreme service conditions.

3.3.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) AC Power (Single Phase) 115V, Category "B" 100 VA

(2) DC Power 28V, Category "B" 2 Amps

3.3.12.1 Additional Input Power - This equipment shall also require the following input power which is not available directly as primary power from the aircraft but must be converted from primary power or provided from sources within the equipment itself.

(1) AC Power (Single Phase) 26V, Category "B"

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.3.12.2 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.3.13 Standby Provisions - The equipment shall provide for a standby position whereby rated voltage shall be applied to all circuits except the integrators.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The equipment shall accept electrical inputs of the ground speed, drift angle, magnetic heading, true air speed and radar fix; manual inputs of present position, target coordinates, magnetic variation, wind speed and wind direction; shall continuously compute and display the present position in latitude and longitude and shall compute the distance and relative bearing from the present position to the destination. The equipment shall also provide an electrical output of true heading, true bearing, drift angle, ground speed and course steering error. The computation of distance and course to destination shall be a planar solution for distances between zero and 200 nautical miles and a great circle solution for distances between 200 and 2000 nautical miles.

3.4.2 Accuracy - The accuracy of the computations performed by the equipment shall be as shown in TABLES 3 and 4. The stated accuracies shall be maintained with no adjustments and shall be valid for all service conditions specified in paragraph 3.3.10.

3.4.3 Input Signals - The equipment shall accept electrical and manual input signals as shown in TABLES 1 and 2.

3.4.4 Output Signals - The equipment shall provide electrical and visual outputs as listed in TABLES 3 and 4.

3.4.5 Modes of Operation - The equipment shall have the following modes of operations.

3.4.5.1 Normal Mode - The equipment shall accept inputs of ground speed, drift angle, magnetic heading and true air speed. The True Ground Speed vector shall be used for dead reckoning and in addition shall be added to the True Air Velocity vector to provide a continuous solution of Wind Velocity. All system outputs shall be available during this mode.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:35Z
Check the source to verify that this is the currentversion before use.

3.4.5.2 Wind Memory Mode - The Wind Memory Mode shall be automatically actuated when the velocity sensor quality signal indicates a temporary loss of velocity sensor operation. The Wind Memory portion of the Computer Set shall then retain the last value of the wind vector computed when the sensor system was operating properly to combine with the current Airspeed vector to form a new Ground Speed vector to be used for dead reckoning computations. All outputs as described in TABLES 3 and 4 shall be available in this mode of operation excepting up-to-date wind as described in TABLE 4.

3.4.5.3 Air Mass Mode - When the velocity sensor system is completely inoperative, manually inserted wind settings based on meterclogical data shall be added to the Airspeed vector to solve for the Ground Speed vector. The equipment instrumentation shall be identical to that used in the Wind Memory Mode and thus provide the same outputs as in the Wind Memory Mode.

3.4.6 Position Up-Dating -

3.4.6.1 Automatic Radar Fix - The equipment shall provide outputs of true bearing and ground distance to a check point, as selected, as either Target No. 1 or Target No. 2, suitable for automatic positioning of search radar azimuth and range cursors (cross hair laying). In addition a signal shall be provided when this check point is within 150 nautical miles from present position. An a-c signal on either of two leads with respect to ground (North-South and East-West) shall cause the appropriate present position counter indication and velocity determined by the phase and magnitude of the signal. The presence of a +28 V dc signal shall be required on a third line with respect to ground to enable the counter slewing. AC and DC excitation and a 26 V ac phase reference shall be provided by the associated equipment.

3.4.6.2 Visual Fix - The equipment shall be capable of inhibiting the accumulation of present position to permit the insertion of visual check point coordinates on the present position counters for a visual position fix up-dating procedure.

3.4.6.3 Position Changes - It shall be possible to change the settings on the Present Position and Target counters while the Computer Set is in operation. Changing the setting of the Target counters shall not interrupt or distrub the computation of present position. Steering outputs for one target shall be continuously available while changing the other target coordinates.

3.4.7 Self-Test - The equipment shall include a front panel mounted test switch permitting integrity testing of the Computer Set without requiring power application to associated subsystems. A standard problem shall be applied to the system to enable verification of operation of the wind computer as well as the present position computation by monitoring the various Computer-Control Panel Counters.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:35Z
Check the source to verify that this is the current version before use.

TABLE 1

ELECTRICAL INPUT SIGNAL CHARACTERISTICS

Input	Range	Synchro Type (Form)	Scale Factor	Zero	Maximum Rate of Change
Ground Speed	50 to 1000 kts	Synchro Trans. 26V 400C	36°/100 kts	Synchro Zero at 550 kts	20 kts/sec max
Drift Angle	180° right to 180° left	Synchro Trans. 26V 400C	1°/1°	Synchro Zero at 0°	7°/min
True Air Speed	50 to 1000 kts	Synchro Trans. 26V 400C	36°/100 kts	Synchro Zero at 50 kts	400 kts/sec max
Magnetic Heading	0° to 360°	Synchro Trans. 26V 400C	1°/1°	Synchro Zero at 0°	20°/sec max
Velocity Sensor Quality	On or Off	D.C. Voltage	28V±1V dc	Note 1.	
Radar Fix		A.C. Voltage	400 Cycles 28V		6048±360 kts

Note 1. 28 V dc for Doppler Reliability and Zero (0) Volts for unreliability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

TABLE 2

MANUAL INPUT CHARACTERISTICS

Input	Characteristics
Present Position	Latitude 80°00'N to 80°00'S Longitude 180°00'E to 180°00'W
Target Coordinates	Latitude 80°00'N to 80°00'S Longitude 180°00'E to 180°00'W
Magnetic Variation	180°E to 180°W
Wind Speed	0 to 250 kts
Wind Direction	0°to 360°

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

TABLE 3

ELECTRICAL OUTPUT SIGNAL CHARACTERISTICS

Outputs	Range	Synchro Type	Scale Factor	Zero	Accuracy
Present Position	Lat. 80°N-80°S Long. 180°E-180°W	Bendix AY-301-S-16 or equivalent 26V 400 Cycle 35:1	1°/1°	Synchro Zero at 0° Lat. and 0° Long.	Less than 1nm/60 min or "P" x distance travel whichever is greater Note 2.
Distance to Target	0 to 2000 nm	Bendix AY-503F-21 or equivalent for each decimal digit	26V, 400 cps	Zeroed at Zero Digit	±1+0.5% 0-200 nm ±5+1% 200-2000 nm
Relative Bearing to Destination	0° to 360°	Bendix AY-503F-21 positioned to true bearing and two AY-5331-25 positioned to true heading	1°/1° 26V, 400 cps	Zeroed at 0°	±1°
True Heading	0° to 360°	Bendix AY-503F-21 or equivalent	1°/1° 26V, 400 cps	Zeroed at 0° True Heading	±0.25°
True Bearing	0° to 360°	Bendix AY-503F-21 or equivalent	1°/1° 26V, 400 cps	Zeroed at 0° True Bearing	±0.75° not to exceed ±3.00 at 20 nm range to target
Ground Speed		2-Pot Output	36°/100 kts		±4%
Course Steering Error	±180°	Bendix AY-503F-21 or equivalent positioned to True Bearing and a AY-533-25 or equivalent positioned to ground track	1°/1°	Zeroed at 0° steering error	±1°

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

TABLE 3

(Continued)

Outputs	Range	Synchro Type	Scale Factor	Zero	Accuracy
Drift Angle	180° left to 180° right	Bendix type AY-503F-21 or equivalent	1°/1° 26V, 400 cps	Zeroed at zero (0) degrees	±0.25°
Radar Ground Range	0-150 nm		2 1/4°/1 nm 2 watts		Linear Potentiometer 5000±5% OHMS Linearity 0.25% Resolution 0.1%
Change in East-West and North-South Distances		Bendix type AY-503-26 or equivalent	2 nm/1° 26V, 400 cps		±0.25%
One Hundred and Fifty-Mile Signal	On-OFF Note 1.		28V, d.c.		±5 nm

Note 1. The equipment shall provide a 28V, d.c. signal whenever the distance from present position to the selected target is greater than 150 nm.

Note 2. "P" as defined in Table 4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

TABLE 4

VISUAL OUTPUT CHARACTERISTICS

Output	Range	Form	Accuracy
Present Position	Lat. 80°N-80°S Long. 180°E-180°W	Counter	Less than 1 nm/60 min or "P" x distance traveled which ever is greater P=±0.25% for latitude computation P=±0.25% for longitude computation from the equator to 60°N or 60°S P=±0.5% for longitude computation for 60°to 70° N or S latitude P=±1.0% for longitude computation for 70° to 80° N or S latitude
Wind Velocity	0 to 250 kts	Counter	± 3 kts
Wind Direction	0° to 360°	Counter	± 1.4°/200 kts ± 5°/25 kts

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.4.8 Wind Averaging - Computed wind speed and direction shall be displayed on the Control CP-824/ASN-66. The display wind shall be a continuous average of the instantaneous wind. Wind averaging is accomplished by the use of large gear trains.

3.5 Detail Requirements -

3.5.1 Computer-Control CP-824/ASN-66 - The computer-control unit shall meet the applicable requirement in Specification MIL-C-6781 for type I control panels and the following requirements.

3.5.1.1 Functions - This unit shall be the cockpit control panel unit, providing all manual controls and counter indicators for the Computer Set. In addition, the portion of the computation instrumentation mechanically associated with these controls and/or counter indicators shall be included in this unit. The control panel dimensions shall be in accordance with MIL-C-6781.

3.5.1.2 Form Factor and Mounting - The Navigational Computer shall conform to the dimensions shown in Figure 1.

3.5.1.3 Weight - The unit shall not exceed 11.75 pounds.

3.5.1.4 Contents of Unit - The unit shall consist of six subassemblies as follows:

(1) Chassis Assembly - This subassembly shall include the unit's main frame, the main wiring harness, lighting panel, target override relays function switch, and an elapsed time indicator.

(2) Longitude Module - This module shall include the various instrumentation associated with present position and target longitudes, including both longitude counters.

(3) Latitude Module - This module shall include the various instrumentation associated with present position and target latitudes, including both latitude counters, the longitude convergence (secant) corrector mechanism, and the scaling transformer.

(4) Wind Module - This module shall include the instrumentation associated with the display of wind information, including the wind velocity and wind direction counters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

(5) Variation Module - This module shall include the instrumentation required for the insertion of magnetic variation information, including the variation counter.

(6) Null Detector - This module shall include the circuitry required to switch the instrumentation to permit a change of destination.

3.5.1.5 Controls -

3.5.1.5.1 Front Panel -

(1) Function Switch - For selection of any of the operational functions or self-test of the Computer Set, the switch shall include a detenting arrangement to prevent inadvertent placing of the switch into the OFF, or TEST positions. The function switch shall have two discrete axial positions when TARGET 1 or TARGET 2 is selected. These positions shall be selected by pushing in or pulling out of the knob. The switch positions are as follows:

(a) Test - Simulated inputs of Air Speed, Ground Speed, Drift Angle and Heading shall be applied to the input servo mechanisms to permit integrity self-testing of the computer.

(b) Off - System shall be inoperative.

(c) Standby - Shall suspend present position computation. The target counters shall read the stored coordinates of TARGET 1.

(d) TARGET 1 - The target counters shall read the stored coordinates of TARGET 1 when the knob is in the normal (in) position. When the knob is pulled out the target counters may be used to insert new target coordinates into TARGET 1 storage.

(e) TARGET 2 - Same conditions as TARGET 1 above, except applies to TARGET 2 reading and storage.

(2) Steering Switch - Shall be a two position toggle switch for selection of TGT 1 or TGT 2 for steering. The switch shall determine the target to which all range and bearing computations are made.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:35Z
Check the source to verify that this is the current version before use.

(3) Position LAT and LONG Knobs - These knobs shall provide for manual insertion of initial present position coordinates and simultaneously set the position counters for visual reference. Knobs shall be used in visual position fix procedure and in-flight adjustment of these knobs shall be permissible.

(4) Target LAT and LONG Knobs - These knobs shall provide for manual insertion of target coordinates and simultaneously set the target counters for visual reference. It shall be possible to change targets during flight without disturbing system operation.

(5) VARIATION Knob - This knob shall provide for manual insertion for magnetic variation information and simultaneously set the variation counter for visual reference.

(6) WIND Knobs - These knobs shall provide for manual insertion of Wind Direction and Velocity. The knobs shall simultaneously set the wind counters for visual reference.

3.5.1.5.2 Rear Cover Access -

(1) Interchangeability - Two screwdriver adjustment potentiometer controls shall be labeled SR SCALE and LR SCALE to achieve interchangeability in the course and distance section of the Computer Set. SR refers to the short range computation, while LR identifies the long range instrumentation.

3.5.1.6 Front Panel Displays -

(1) Position LAT and LONG Counters - These counters shall continually change to display the present position of the airplane during flight in degrees and minutes of latitude and longitude, in one minute increments.

(2) Target LAT and LONG Counters - These counters shall display target coordinates in degrees and minutes of latitude and longitude in one minute increments.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

(3) WIND FROM DEG TRUE AND KNOTS Counters - These counters shall continually display current wind direction and wind velocity information, computed during normal mode operation. Wind Direction shall be displayed through 360 degrees (true) in one degree increments; Wind velocity shall be displayed from zero (0) to 250 knots in one knot increments.

(4) VARIATION Counter - This counter shall display magnetic variation settings from 180 degrees East thru zero to 180 degrees West in one-tenth degree increments.

3.5.1.7 Electrical Connections - Connections to external circuits shall be provided as shown in Figure 2.

3.5.2 Amplifier-Computer AM-4341/ASN-66 - The amplifier computer unit shall meet the following requirements:

3.5.2.1 Function - This unit shall include all of the computer electronic circuit cards, power supply, and all electro-mechanical computer mechanism not associated with front panel controls and/or displays. This unit shall provide required signal amplification as well as various computational tasks.

3.5.2.2 Form Factor and Mounting - The Amplifier Computer shall be enclosed in a case and shall conform to the dimensions shown in Figure 3.

3.5.2.3 Weight - This unit shall not exceed 24.25 pounds.

3.5.2.4 Contents of Unit - The unit shall consist of the following assemblies:

(1) Chassis Assembly - This subassembly shall include the units main frame and chassis, main wiring harness, electro-mechanical relays, crystal oscillator, and an elapsed time indicator.

(2) Range Module - This module shall include the various instrumentation associated with great circle distance to destination.

(3) Bearing Module - This module shall include the various instrumentation associated with great circle bearing to destination.

(4) Velocity Module - This module shall include the instrumentation associated with Ground Speed and Airspeed information.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

(5) Ground Track Module - This module shall include the instrumentation associated with Drift Angle, True Heading and Ground Track information, as well as providing outputs of changes in North-South and East-West distances.

(6) Power Supply - This module shall provide the various bias voltages required for system operation.

(7) Memory Modules - Memory modules shall provide for storage of target coordinates.

(8) Step Motor Drive - Two circuit cards shall contain the required logic isolation and amplification to power three system step motors.

(9) Integrator Cards - The integrator circuitry shall consist of a frequency converter card, binary divider card, resolver drive card, and zero cross-ind detector card required for the precise computation of distance traveled from the velocity input. In addition, two cards shall be associated with functions related to radar updating.

3.5.2.5 Electrical Connections - Connections to external circuits shall be provided as shown in Figure 2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Tests

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:35Z
Check the source to verify that this is the currentversion before use.

selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper correction action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run in Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run in Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workman ship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:35Z
Check the source to verify that this is the currentversion before use.

occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall be conducted only when Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

* NOTE: If by contractual action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Sampling Phase shall be conducted.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan I shall be used.

4.4.3.2 Reliability Sampling Phase Tests - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the section entitled "Sampling Phase of Product on Reliability Tests". For the Sampling Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan III shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.4.3.2.1 Lot Size for Sampling Phase - The equipments constructed during one month shall be one lot.

4.4.3.3 Test Details - The test details such as the length of the Test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase or the Reliability Assurance Tests.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:35Z
Check the source to verify that this is the current version before use.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:35Z
Check the source to verify that this is the current version before use.

Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal Room
Altitude	Normal Ground (0-5000 ft)
Humidity	Room Ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. Voltage	27.5 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep the daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of past failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all test failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:35Z
Check the source to verify that this is the current version before use.

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or completed equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:35Z
Check the source to verify that this is the currentversion before use.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - Provide navigation in naval aircraft.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:35Z
Check the source to verify that this is the current version before use.

in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Non-Repairable Subassemblies - As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.6 The parentheses (*) when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Asterisk-In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.8 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment:

Item	Type
Navigation Set, Radar	AN/APN-153(V)
Compass System	MF-1
Air Data Computer	128SCAV400-1 (Grumman) or equal
Analog to Digital Converter	128SCAV402-1 (Grumman) or equal
Cursor Computer	128SCAV403-1 (Grumman) or equal
Course Attitude Data Group	AN/ASA-51
Horizontal Situation Indicator	128SCAV166-5 (Grumman) or equal
Map Display	HDV-1H (ACF)
Antenna Position Indicator (BDHI Modified)	128SCAV407-1 (Grumman) or equal

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

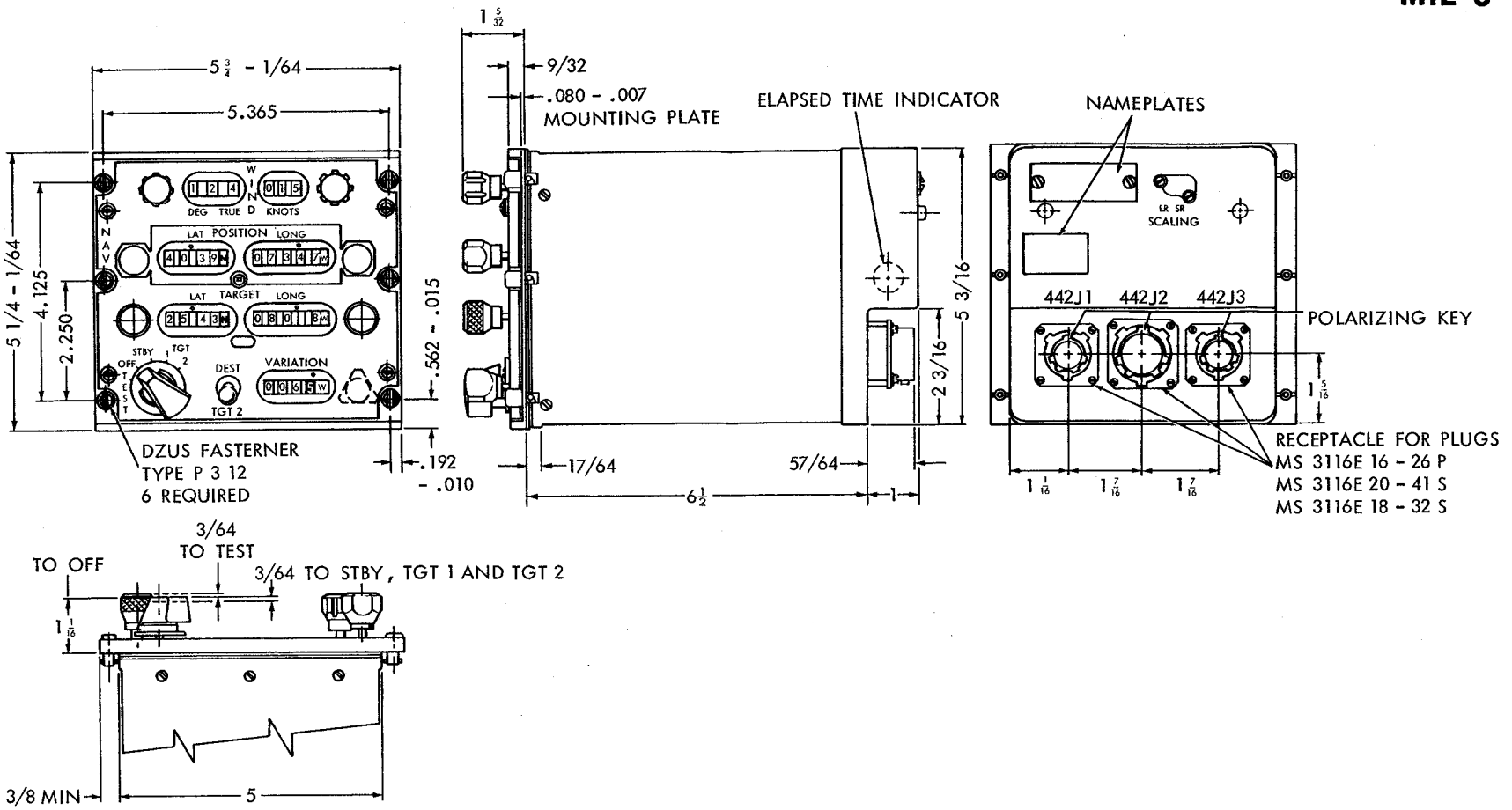
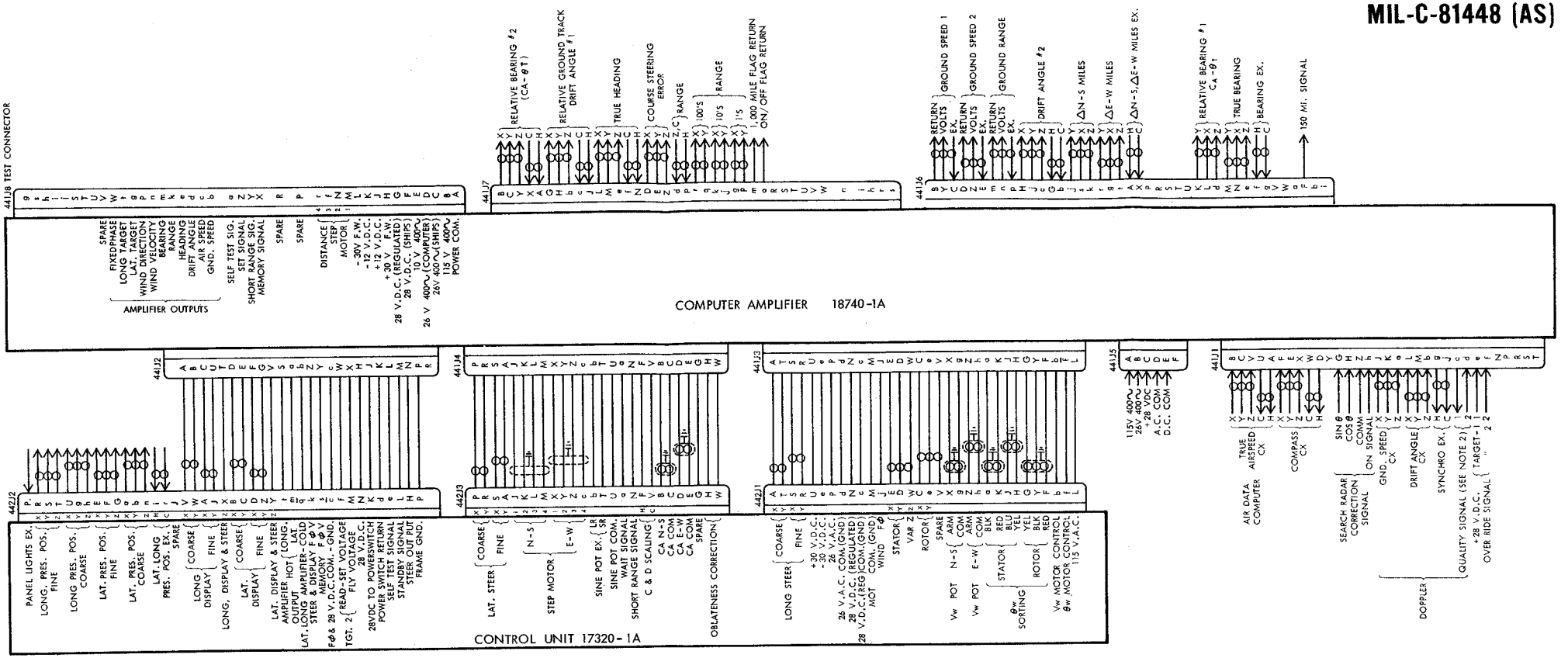


FIGURE 1

COMPUTER-CONTROL CP-824/ASN-66

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.



NOTE:

1. EXCEPT AS HERE LISTED, ALL ARINC CONNECTIONS (X, Y, Z, H & C) HAVE 0° INDEX REFERENCE POSITION & POSITIVE ROTATIONAL REFERENCE FOR INCREASING FUNCTION OR DIAL READING.

FUNCTION	INDEX REF. POSITION	ROTATIONAL REFERENCE
DRIFT ANGLE CX	0°	POSITIVE FOR DRIFT TO THE RIGHT OF MAGNETIC HEADING
GROUND SPEED CX	550 KNOTS	POSITIVE FOR INCREASING GROUND SPEED
TRUE AIR SPEED CX	50 KNOTS	POSITIVE FOR INCREASING TRUE AIR SPEED
ΔE-W MILES	0°	POSITIVE FOR INCREASING EAST
ΔN-S MILES	0°	POSITIVE FOR INCREASING NORTH

CODE

CX = CONTROL TRANSMITTER

CONNECTOR	MATES WITH
441J1	MS3116 E-18-32P
441J2	MS3116 E-16-26P
441J3	MS3116 E-18-32PW
441J4	MS3116 E-16-26SW
441J5	MS3116 E-10-06S
441J6	MS3116 E-20-41S
441J7	MS3116 E-20-41SX
441J8	MS3116 E-20-41PW
442J1	MS3116 E-18-32S
442J2	MS3116 E-20-41S
442J3	MS3116 E-16-26P

2. DOPPLER QUALITY SIGNAL
COMPUTER IS IN DOPPLER MODE OF OPERATION WHEN 28 VDC IS APPLIES TO PIN g ON CONNECTOR 441J1

FIGURE 2

INTERCONNECTION DIAGRAM AN/ASN-66

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

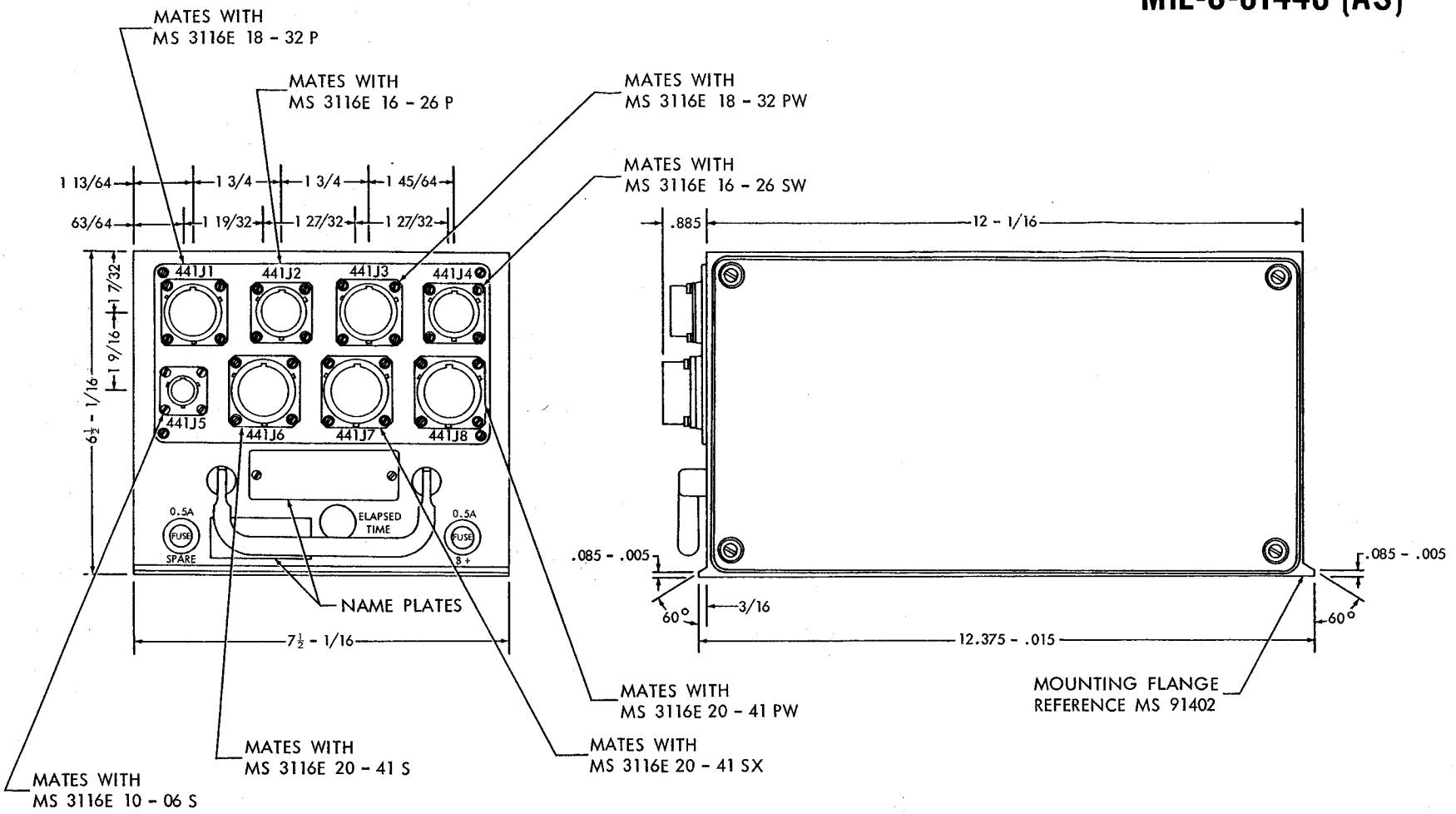


FIGURE 3

AMPLIFIER - COMPUTER AM-4341/ASN-66

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-C-81448(AS) COMPUTER SET, NAVIGATION AN/aSN-66(*)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

C-8279

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.